NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SECTION A : 5 marks each

1. Divide 10 by $\frac{1}{2}$ and add 10.

Solution

$$\begin{aligned} 10 \div \frac{1}{2} &= 20 \\ 20 + 10 &= 30 \end{aligned}$$

2. While using a skipping rope, Anita jumps 52 times in one minute. Jumping at this same rate, how many times does Anita jump in 75 seconds?

Solution

52 times in 60 sec

$\frac{52}{60}$ in one sec

$\frac{52}{60} \times 75$ in 75 sec $= 65$ times

3. A public holiday is always celebrated on the third Wednesday of a certain month. In that month, the public holiday cannot occur on the 22nd. Why?

Solution

first Wed can fall on	1	2	3	4	5	6	7	of the month
2nd Wed	1+7	2+7	3+7	...	...	...	7+7=14	
3rd Wed	1+7+7	2+7+7	3+7+7	...	...	...	7+7+7=21	

Therefore cannot be on 22nd.

4. Robinah has 4 tasks to do and Sandra has 14 tasks to do. In order for Robinah and Sandra to do the same number of tasks, how many of Sandra's tasks should Robinah do?

Solution

Between them, $4 + 14 = 18$ tasks

Same tasks for each $18 \div 2 = 9$

So Robinah has to do extra $9 - 4 = 5$ tasks from Sandra.

5. Maggie and Rosie are 90 metres apart. Starting at the same time, they run towards each other. Maggie runs twice as fast as Rosie. How far has Maggie run when they meet?

Solution

Maggie	Rossie
$2x$	x

$$\begin{aligned} 3x &= 90 \\ x &= 30 \end{aligned}$$

Maggie has run 60m

6. In a family of five children, Dan is older than Ben. Ben is older than Eva. Eva is younger than Annet. Annet is younger than Ben. Ben is younger than Cathy. Who is the third oldest?

Solution

Annet	Ben	Cathy	Dan	Eva
A	B	C	D	E
$D > B$	$B > E$	$A > E$	$B > A$	$C > A$

So D and C are older than B
E and A are younger than B
So 2 people are older than B and 2 people are younger than B,so 3rd oldest must be B (Ben)

7. Study this pattern and find $6 + 6$.

$$\begin{aligned} 2 + 2 &= 44 \\ 3 + 3 &= 96 \\ 4 + 4 &= 168 \\ 5 + 5 &= 2510 \end{aligned}$$

Solution

$$\begin{aligned} a + b &= (a^2)(a + b) \\ 6 + 6 &= (6^2)(6 + 6) \\ &= 3612 \end{aligned}$$

8. If 2.2×10^{25} is doubled, what is the result? (Do not simplify)

Solution

$4 \cdot 4 \times 10^{25}$ (only double $2 \cdot 2$)

9. When a container is $\frac{1}{2}$ full, it contains exactly enough water to fill three identical glasses. How full would the container be if it had exactly enough water to fill four of the same glasses?

Solution

10 glasses are same as $\frac{1}{2}$ of containr.

1 glass is the same as $\frac{1}{6}$ of container.

4 glasses are the same as $4 \times \frac{1}{6} = \frac{2}{3}$ of container.

10. A bank employee is filling an empty Automatic Teller Machine (ATM) with bundles of Shs. 5,000, 10,000 and 20,000 bills. Each bundle has 100 bills in it and the machine holds 10 bundles of each type. What amount of money is required to fill the machine?

Solution

1 bundle of shs 5,000 is worth 500,000

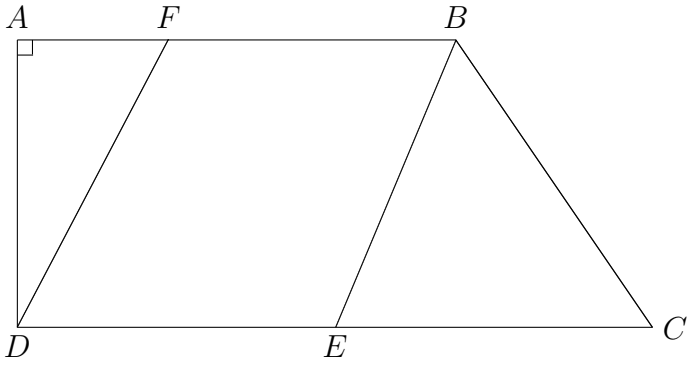
1 bundle of shs 10,000 is worth 1,000,000

1 bundle of shs 20,000 is worth 2,000,000

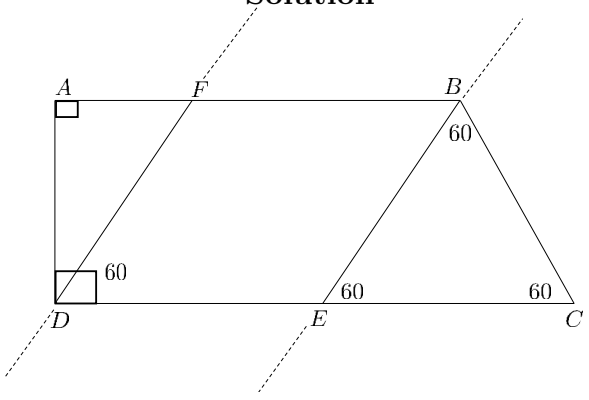
10 bundles of each type gives $10(5000,000 + 1,000,000 + 2,000,000) = 35,000,000$

SECTION B : 10 **marks each**

11. In trapezoid $ABCD$, AB is parallel to DC and angle DAF is 90 degrees. Point E is on DC so that $EB = BC = CE$. Point F is on AB so that DF is parallel to EB. Find angle FDA in degrees.



Solution



$DF \parallel BE \quad \angle FDC = \angle BEC = 60^\circ$

$AB \parallel \ to \ DC \ \ AD \perp AB$

AD is also $\perp$ to DC

$$\angle ADC = 90^\circ$$

$$\begin{aligned} \text{Angle } FDA &= 90^\circ - 60^\circ \\ &= 30^\circ \end{aligned}$$

12. Allen, Betty and Charles play a two-player game that never ends in a tie. In a recent tournament between the three players, a total of 60 games were played and each pair of players played the same number of games.
    - When Allen and Betty played, Allen won 20% of the games.
    - When Betty and Charles played, Betty won 60% of the games.
    - When Charles and Allen played, Charles won 40% of the games.

    How many games did Betty win?

Solution

Each pair played 20 games

Allen Vs Betty: Allen won 20% of 20 ie 4 wins, therefore, Betty won 16 games.

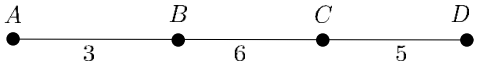
Betty Vs Charles: Betty won 60% $60\% \times 20 = 12$ wins.

Therefore, Betty won $12 + 16 = 28$ games.

13. Line segment AD is divided into three segments by points B and C so that $AB : BC = 1 : 2$ and $BC : CD = 6 : 5$. The length of AD is 70 cm. What is the length of AB?



Solution

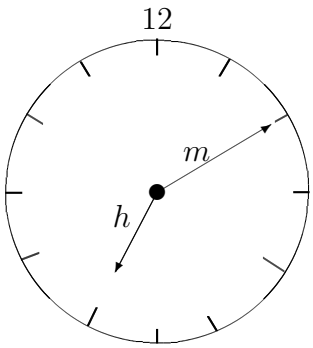


With given ratios

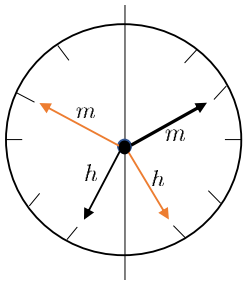
$$\begin{aligned} AB &= 3 \\ BC &= 6 \\ CD &= 5 \\ 3+6+5 &= 14 \quad but \\ AD &= 70cm \end{aligned}$$

$$\text{Therefore, } AB = \frac{3}{14} \times 70 = 15cm$$

14. Peter sits on a chair in a room. Behind him is a clock. In front of him is a mirror. In the mirror, he sees the image of the clock as shown. What is the actual time?(note: h is the hour hand and m is the minute hand).



Solution



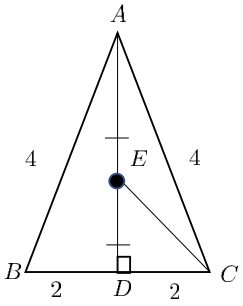
$\simeq$ time 4 : 50 reflect both hands and read off the the time

15. Each side of Equilateral triangle ABC has length of 4 cm. The midpoint of BC is D, and the midpoint of AD is E.

    (a) Sketch the equilateral triangle and indicate all the points.

    (b) Find the length of EC.

Solution



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\begin{aligned}
AD \perp BC \;\; AB &= AC, \;\; D \text{ is mid print of } BC \\
(AC)^2 &= (AD)^2 + (DC)^2 \\
\text{So } (AD)^2 &= 16 - 4 \\
&= 12cm \\
(EC)^2 &= (ED)^2 + (DC)^2 \\
&= (ED)^2 + 2^2 \\
\text{Since } (ED) &= \frac{1}{2}AD \\
(ED)^2 &= \frac{1}{4}(AD)^2 \\
&= 3 \\
(EC)^2 &= 3 + 2^2 \\
&= 7cm \\
EC &= \sqrt{7}cm
\end{aligned}
$$